

TREATMENT OF PHOBIAS IN A HOSPITALIZED CHILD

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Summary—A 12-year-old girl, hospitalized on a psychiatric intensive care unit, was treated for phobias of school and unfamiliar males. Treatment focused on decreasing several avoidance responses and increasing prosocial approach responses. Treatment consisted of instructions, performance feedback, participant modeling and social reinforcement. Marked changes were evident when treatment was introduced in a multiple-baseline design across several phobic and prosocial behaviors. Treatment effects generalized beyond the persons and situations included in training, were reflected in overall global ratings by persons unfamiliar with the treatment, and were maintained up to a 7-week follow-up when the child returned to the hospital for reassessment. Contact with the child 21 weeks after discharge indicated that the gains were reflected in school attendance and social interaction in everyday situations.

Intense fear is frequent in children, with an estimated prevalence of approximately 8% of the total population (Crewe, 1973; Miller, Barrett and Hampe, 1974). Although many childhood fears improve over the course of development, the more intense ones may persist (Agras, Chapin and Oliveau, 1972). Phobias result in distress and decreased adaptability, occasionally are resistant to change and necessitate therapeutic intervention (Ollendick, 1979). Several attempts have been made to classify childhood fears (Miller *et al.*, 1972; Scherer and Nakamura, 1968). Angelino, Dollin and Mech (1956) grouped into broad categories such as safety, school and social contacts those areas in which children are most likely to be phobic.

The treatment of phobias in children has a long tradition of behavior modification beginning with the work of Watson and Rayner (1920) and Jones (1924). Among the treatments employed, the bulk of the research has focused on systematic desensitization (Bentler, 1962; Eysenck and Rachman, 1965; Lazarus, 1960; Weber, 1936; Weinstein, 1976), modeling (Bandura, Blanchard and Ritter, 1969; Hill, Liebert and Mott, 1968; Kornhaber and Schroeder, 1975; Lewis, 1974; Ritter, 1968) and reinforced practice (Holmes, 1936; Luiselli, 1977; 1978; Pomerantz *et al.*, 1977).

A notable omission in this area of research has been the development of appropriate positive or approach behaviors once phobias have been diminished. Elimination of avoidance behaviors

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does not necessarily insure that approach responses are achieved. The present study evaluated whether reductions of separate phobias were followed with increases in appropriate approach behaviors. Approach behaviors were developed directly if they did not occur following initial treatment. The child who received treatment suffered intense fears which led to sufficient dysfunction to precipitate hospitalization. Typically, treatments for phobic children are of relatively short duration (e.g. 4–5 sessions) and focus on one or a few responses (Ollendick, 1979). Because of the severity of the case described below, more extended treatment across a larger range of behaviors was investigated.

METHOD

Subject and setting

The subject was a 12½-year-old white female inpatient of an 18-bed children's psychiatric care unit at the University of Pittsburgh School of Medicine. Her DSM III diagnosis was Anxiety Disorder of Childhood; School Refusal and Social Phobia. Her primary problems included an extreme fear of all unfamiliar men, which occurred after alleged sexual molestation by her grandfather, and school refusal due to extreme fear and anxiety in the school setting. Behaviors associated with these two problems included avoidance of school and unfamiliar men, diminished ability to interact with peers, and multiple somatic complaints. Prior to admission, the child was enrolled in a sixth grade elementary classroom. She was of normal intelligence (full scale WISC-R = 95) and gave informed consent to participate. The study was conducted in a room, 3½ × 5 m, furnished with a table and three chairs on the inpatient unit. A one-way mirror separated the room and an adjacent observation room with videotape equipment. All data were obtained in this setting.

Assessment

Assessments were made based on responses to role-play situations presented by the therapist (first year resident in psychiatry) and answers to questions posed by unfamiliar males. Throughout the study 10 role-play situations were used each session. Five of these pertained to unfamiliar males; these included asking a man for a donation to a children's hospital, asking a salesman to try on a pair of shoes, meeting a new male therapist, welcoming a peer's father to the psychiatric unit, and sitting next to an usher at a wedding reception dinner. Anxiety-provoking school situations were picking up a graded semester report with a poor mark, being excluded by peers during an art project, speaking in front of a class, being accused of cheating by the teacher, and being sent to the principal because of tardiness. The presentation of these scenes required approximately 15 min.

The presentation of each scene was in a standardized

fashion. Initially the therapist would read the description of the situation (narrator), then she would assume the role of role-model prompt (i.e. teacher, peer, sales person, etc.) and the subject would respond. A verbal exchange continued from 1 to 3 exchanges on the part of both the therapist and subject, depending on the type of scene presented and the child's response. Each role-play was acted out to enhance the realism of training. For example, when the child engaged in the scene on speaking in front of the class, she was required to speak in front of the therapist.

In each session, direct observations were made of several inappropriate (avoidance) and appropriate (pro-social) behaviors pertaining to the child's school fears and fear of unfamiliar men.

Inappropriate behaviors included:

Stiffness. Lack of mobility in the trunk and limbs while responding to the therapist. Specific behaviors targeted included shoulders raised and hunched, neck stationary, no movement of head, lack of trunk movement and upper arms held tight and close to the body. This behavior was rated on a 3-point scale (1 = none, 3 = very).

Nervous mannerisms. This behavior was based on facial features, hand, torso and arm movements. Halting speech, low voice volume, nervous giggling, furrowing of the brow and grimacing characterized this behavior. Nervous mannerisms were rated on a 5-point scale (1 = not at all, 5 = very much). Adjectives that characterize this behavior were "fidgety" and "being flustered".

Self-rating of anxiety. The experimenter asked the child to rate on a 3-point scale from 1 (not at all) to 3 (very much) how nervous she felt during the scene. This was recorded by the experimenter.

Appropriate behaviors included:

Eye contact. Eye contact was defined as the subject's eyes looking at the experimenter's eyes during conversations and was measured on a 3-point scale (1 = few glances, 3 = glances most of the time).

Appropriate affect. This involved the child smiling during conversations when smiling was appropriate to the content, i.e. greeting parents, meeting strangers, etc. It also involved the use of full and lively voice intonation, initiations of further conversation and being appropriately assertive but not demanding. Appropriate affect was rated on a 3-point scale (1 = inappropriate, 3 = appropriate).

Appropriate body movement. Appropriate body movement was defined as approaching the experimenter when speaking, standing oriented towards the experimenter, head facing the experimenter and using hand gestures that fit the conversation. The behavior was rated on a 3-point scale (1 = inappropriate, 3 = appropriate).

Overall social skills. Social skill was defined as the actual verbal content of a response to a scene with respect to both relevance to the scene and overall social acceptability. Additionally, nonverbal behaviors such as gestures, smiling and related responses were included. The behavior was rated on a 5-point scale (1 = inappropriate, 5 = very appropriate).

Probe assessment. In addition to the daily assessment of the above behaviors, two types of probe assessments were made. The first probe was conducted at sessions 3, 5, 8, 14 and 23. The purpose of this was to evaluate whether treatment effects extended to an unfamiliar male who only had contact with the child during these special

assessment periods. During this probe assessment, the scenes were presented as noted above. Stiffness, nervous mannerisms, self-report, eye contact, appropriate affect, appropriate body movement, and overall social skills were rated during these sessions, as described above.

During the same probe assessment occasions, the child was interviewed by one of three male faculty members unfamiliar to her, in a rotating sequence. The males were dressed in suits and introduced themselves as doctors at the hospital. Each male asked the following questions: (1) Can you tell me about yourself?, (2) What do you do when you have free time?, (3) What kinds of things make you happy?, and (4) What kinds of things make you nervous? While responding, the child was rated for stiffness, nervous mannerisms, eye contact, appropriate affect, body movement, and overall skills, as described above.

Social validation. Ratings of persons not directly involved in treatment were obtained to determine whether changes in the child's behaviors could be readily detected. Such ratings have been recommended and employed to validate or assess socially the clinical importance of therapeutic changes (Kazdin, 1977; Wolf, 1978). Ten direct care staff from the children's unit (registered psychiatric nurses and baccalaureate level psychologists) rated a segment of the first baseline and final treatment session (2 scenes on school phobias and 2 on unfamiliar males). Staff were those who did not spend excessive amounts of time or individual contact with the child on the unit. The order of tape presentation was varied so that half the staff viewed baseline and the other half viewed the treatment session first. Staff were blind to the order of the tape presentation. For each of the scenes staff rated the child on three dimensions related to discomfort (nervousness, meekness, and uncomfortableness) and three dimensions related to social presentation (security, sociability and openness). The six ratings were defined as follows:

Nervousness. This behavior was defined as excessive fidgeting, hesitation in speech, low voice volume and inappropriate giggling.

Meekness. This behavior was defined as lack of verbalizations or verbal expressiveness. Meekness consisted of failure to express appropriate assertiveness in situations where voicing one's opinion would be desirable.

Uncomfortableness. Uncomfortableness was defined as evincing tension in both verbal, affective and motoric responding. Thus, behaviors such as pressured speech, inappropriate smiling, uneasy facial and bodily posturing constituted this behavior.

Security. This behavior was defined as social behavior which made the child appear to be overtly more confident and self-assured. Behaviors were both verbal and nonverbal and involved looking at the respondent rather than down or away while speaking, speaking at an audible level and responding quickly rather than in a slow halting gait.

Sociability. Sociability was defined as having a friendly, amiable appearance and appropriate verbal content of speech.

Openness. This behavior was defined as being receptive, interested and willing to express one's feelings.

Follow-up

Seven weeks after the child completed treatment and was discharged, she returned to the hospital for follow-up assessment. Assessment included the direct observations of the 10 role-play situations and the second probe assessment involving an interview by one of the doctors. Because the child's home was far away from the treatment setting, she could not return conveniently for repeated follow-up assessment. However, the family was contacted 21 weeks after discharge for reports on how well she was doing.

Raters and observer agreement

Raters were two undergraduate psychology students. They were trained to a reliability of 0.80 or above on stiffness, nervous mannerisms, overall social skill, eye contact, appropriate affect, and appropriate movement. This training consisted of a discussion of definitions of target behaviors, examples of these problems and conducting practice rating sessions. Training of raters required 15 hr. Reliability was assessed by obtaining a sum for each rater across the 10 scenes separately for each of 6 behaviors. Pearson correlations were calculated for each of the behaviors ranging from $r = 0.81$ to 0.98 (mean $r = 0.92$).

Experimental design

Baseline. This condition consisted of a daily assessment described previously. No training was provided. Probe measures were also taken periodically during this condition.

Treatment. Training was based on participant modeling (Bandura, 1969; Matson, 1981). At the beginning of each session, 5 min were spent in a general discussion of fears and phobias and how training could be used in the natural environment. This was followed by training. Training was oriented around the scenes described in the assessment section and consisted of two components. In the first, the therapist modeled appropriate behaviors in the situation complete with each target response which was identified by the therapist. Next, the child was asked to act out the scene described by the therapist with information feedback and modeling of responses being made as the child's performance progressed. Social reinforcement for appropriate behavior was provided and information on how to correct the response was given whenever possible. At the completion of the child's performance, the therapist summarized the assets and deficits in her behavior with respect to the target behaviors. Specific aspects of the performance that were incorrect were modeled and performed a second time. Finally, the entire scene was rehearsed. A new scene was then introduced and the same procedure was followed. Training focused on separate sets of behavior to meet the requirements of a multiple-baseline design.

RESULTS

Inappropriate behaviors were rapidly decreased and appropriate behaviors rapidly increased when treatment was introduced. Figure 1 shows that target behaviors were treated in one of three multiple-baseline lags after 5-14 baseline sessions.

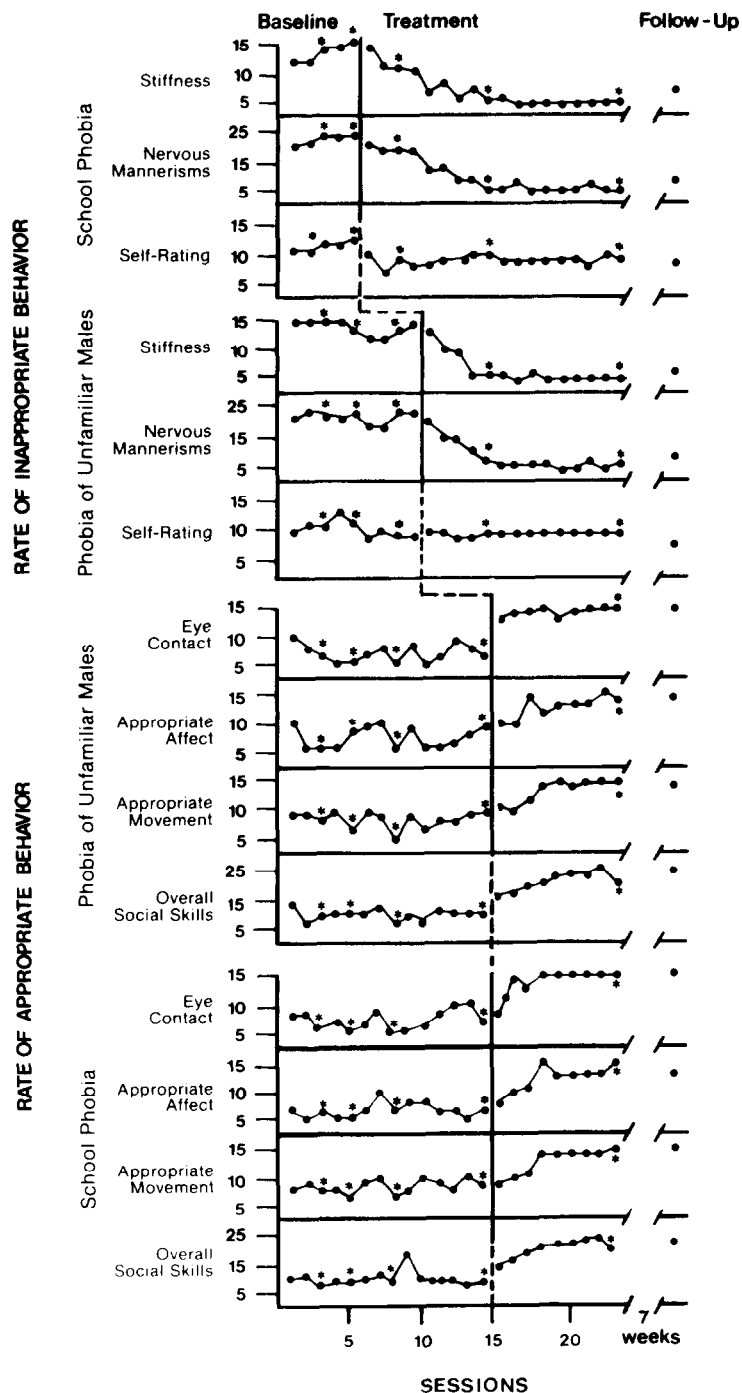


Fig. 1. Rates of appropriate and inappropriate behaviors across experimental conditions on daily assessments and probes (indicated by asterisks) by a second rater.

Two inappropriate behaviors: nervous mannerisms and stiffness were treated first, after a high stable baseline was established. With the advent of treatment, target behaviors were rapidly decreased. Responses attained were maintained at near zero levels for nervous mannerisms and stiffness. Although self-ratings decreased markedly, the effects were not as dramatic as with the two target behaviors.

A similar pattern was seen on a phobia of unfamiliar males, with one exception. Self-ratings decreased with the introduction of treatment on school phobia, and no additional decreases occurred when the phobia of unfamiliar males was treated. All appropriate behaviors remained at low levels even after inappropriate behaviors were decreased, with the exception of appropriate movements for the phobia of unfamiliar males. All appropriate behaviors changed in desired directions with the advent of treatment. Probe assessments evinced changes analogous with data on primary measures of target behaviors.

Figure 2 illustrates the results of probe assessments where the child was interviewed by one of three doctors. Stiffness, nervous mannerisms and social responsiveness changed in positive directions and remained at improved levels with the introduction of treatment. Similarly the generalization of treatment effects to these unfamiliar doctors was exemplified by the gradual improvement of overall social skills as additional target behaviors were treated.

The final measure was an assessment of social validity of treatment effects made by psychiatric staff. Correlated *t*-tests from baseline to treatment were computed across the ratings completed by staff. From the six items rated, three each were combined to comprise discomfort and social presentation scores. Discomfort included the three items of nervousness, meekness and uncomfortableness; social presentation included ratings of security, sociability and openness. From baseline to treatment, significant improvements were evident for both discomfort [$t(9) = 14.08, P < 0.001$] and social presentation [$t(9) = 15.55, P < 0.001$].

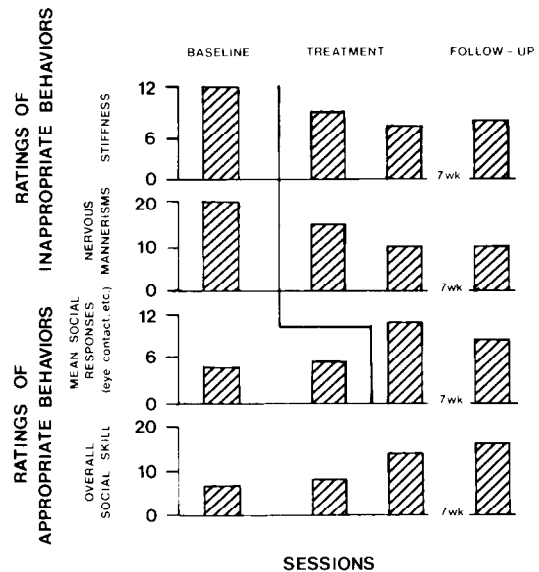


Fig. 2. Rates of appropriate and inappropriate behaviors across experimental conditions on probes by unfamiliar doctors.

Follow-up was evaluated by retesting on the specific target behaviors as well as the interview ratings with one of the doctors. The complete assessment battery was administered after the patient had been discharged. In addition, 21 weeks after discharge the family was contacted by phone to report the child's current status in detail. Follow-up data at the 7-week assessment indicated that the gains were maintained in each area of performance (see Figs 1 and 2). Contact 21 weeks after discharge revealed that the child was doing well in the areas focused on in treatment. Prior to treatment, the child would not leave home without her mother and refused to attend school. These were no longer problems. The child was performing adequately at school and especially well in some areas (e.g. reading) which led to her advancement. Her social skills were also reported as markedly improved. Specifically, she had interacted appropriately and in one instance appropriately assertive with strangers. And, she was reported to be using her skills in everyday situations

such as at stores and on the school bus. Overall, the follow-up data and anecdotal reports suggested that treatment effects were maintained and transferred to everyday situations.

DISCUSSION

In the present study intense fears were rapidly modified in a child hospitalized primarily for these problems. Secondly, adaptive behaviors incompatible with the target behaviors were trained. Appropriate behavior did not increase "spontaneously" with remission of phobic responses. Additionally, the appropriate behaviors were used both in the training situation in other analogue situations (assessment sessions) and on the ward (based on informal reports of staff suggestions).

Interestingly, little generalization occurred across target behaviors for a specific phobia, for the different phobias, or from appropriate to inappropriate behavior. On the other hand, generalization from one assessor to another was observed. This situation was partially expected since behaviors that were hypothesized to be interdependent before training were placed together for treatment. Secondly, based on historical information from the parents, teacher and child herself, a primary cause of the problem was a lack of learned skill in both coping with anxiety and adaptive behavior. Thus, it would not seem surprising that generalization across assessors would occur once the skill was acquired. Additionally, since training was of a much longer duration than typically has been reported in other studies, it is not surprising that skill level reached near optimal proficiency across target behaviors and was maintained at that rate at the 7-week follow-up. The reports of maintenance and generalization 21 weeks after treatment are important considerations in evaluating the treatment. However, systematic assessment of performance in everyday situations was not completed.

Another important issue in the present study was the systematic and broad-based attempt at

assessment. A major concern in many behavioral treatment studies is that the focus is almost exclusively on two, three or four discrete behaviors measured in one or two settings. It is the authors' contention that behavioral research in psychopathology should be broadly based since the standard for labeling such disorders is DSM III. Secondly, behaviorally-oriented researchers and other investigators have for some time stressed the importance of assessing collateral behaviors to give a wider scope to evaluating treatment (Kazdin and Wilson, 1980). Thus, a two-fold rationale for such an approach exists. In the present study multiple appropriate and inappropriate behaviors were assessed with different interviewers, types of measures and raters. This approach not only gives social validity to treatment effects but aids in establishing that the effects are of a clinically significant magnitude.

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